Breaking Into Cyber

With Aggie Butvilas

Hacker @ Zebra Technologies

$ who am i

Name: Agnė "Aggie" **Butvilas**

Job Title: "Information Security Engineer"

About Me:

- Product Security @ Zebra Technologies
- Analog Photographer in Chicago's underground music scene
- Hardware Hacker + Application Security Penetration Tester

Agenda:

1. My Career Journey
2. What my day-to-day life looks like
3. Ways to set *yourself* up for success

How I Got Started

- DePaul University 2020-2024
  - Major in Cybersecurity, Minored in Computer Science
  - Active in **Cybersecurity Competitions**
  - Involved in Cybersecurity Club
  - **Worked in DePaul's Cybersecurity Research Lab**
    - Research on Zebra Projects & Malicious QR Codes

- Began Attending **Cybersecurity Conferences** in 2022
  - Local: Thotcon, BSides 312 & BSides Chicago, Blue Team Con
  - Regional: GRRCon, CypherCon, CornCon, SecretCon
  - The Big Ones: DEFCON, H.O.P.E,
  - Many, many others revolving around different specializations

What I Do At Work

- Leading a project to construct a Hardware Hacking Lab
- Working with vendors/sales reps to integrate new tools
- 3-week long penetration tests
    - 2 weeks of testing, 1 week of writing a report

What I Do Outside of Work

- Burbsec - Cyber Security Meetups on Thursdays
  - https://burbsec.com
  - Different locations across Chicagoland
  - Closest one is Burbsec East
    - Last Thursdays of the Month @ Navigator Taproom in Logan Square

Chicago
Every Last Thursday

Learn More

Wheeling
Every Second Thursday

Learn More

Naperville
Every Third Thursday

Learn More

Schaumburg
Every First Thursday

Learn More

Crystal Lake
Every Fourth Thursday

Learn More

Hickory Hills
Every Second Thursday

Learn More

South bend, IN
Every First Monday

Learn More

Las Vegas
Monthly Meetup

Learn More

Galway, Ireland
Monthly Meetup

Learn More

Career Advice Speedrun

Projects Projects Projects Projects Projects

This is the single best way to stand out on a resume/interview

- **Things I did:**
  - Hardware Hacking: Rooting devices, building hacking tools (DIY usb rubber ducky is an easy + impressive one)
  - Capstone Project: Exploiting Traffic Light Systems with a Flipper Zero
  - Built Discord Bots to keep my Python skills sharp
- **Things I saw other people do:**
  - DIY Electronics: Modular synthesizers
  - Automate Everything! w/ Bash Scripts
  - Homelabs of varying complexity
    - Run your own web server, create a network-wide adblock, host a honeypot to bait malicious hackers

Resume Do’s & Don’ts

Do:

- Include your skills + certifications at the top
- Tailor Your Resume for Each Job You Apply To
    - Have 1 long “master list” resume & cherry pick from there for each job application

Don’t:

- Don’t Over Do It:
    - Keep things to **One Page**
    - **Don’t list irrelevant achievements.** High school accomplishments don’t matter after college :)
    - No need to list *all* your skills, just **skills relevant to the job** you’re applying for

Set Yourself Up For Success Before Graduating

- **Internship AND/OR Research Project**
  - UIC has many research labs, find something you're interested in and get involved!
  - Internships in Security
- **Specialize as Early As Possible**
  - Figure out what skills are relevant to your area of interest & focus on becoming an expert these skills/tools
  - Talk to people working in these specializations already and ask how they got there!

Opportunities Unique For Students

- **Cyber Auto Challenge, Cyber Truck Challenge, Cyber Medical Challenge, Cyber Tractor Challenge, etc…**
  - All-Expenses-Paid, week-long workshops for students focused on a variety of fields
  - https://www.cyberauto-challenge.org
  - https://www.cybermedicalchallenge.org
  - https://www.cybertruckchallenge.org
  - All free!!! Travel, food, lodging, and workshops!
- **CCDC - Collegiate Cybersecurity Defense Competitions**
  - https://www.nationalccdc.org/index.php/competition/competitors/registration
  - You must have at least 4 players (3 players plus a team captain) to register and you must have a staff or faculty member from the institution you are representing to serve as the coach of record.
- **CPTC - Collegiate Penetration Testing Competitions**
  - https://cp.tc

https://www.cybermedicalchallenge.org

I attended Cyber Auto 2023, Cyber Medical 2024, and Cyber Tractor 2024

Questions!

Find Me Online

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